Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/19/2012
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Mapping gravitropic signaling and response: from ions to growth		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/27/2009	**End Date:**	05/26/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/26/2013 per NSSC (Ed., 6/1/2012)		
Key Personnel Changes/Previous PI:	March 2012: The postdoc who was originally hired to work on the project, Dr. Won-Gyu Choi, has moved on to another project and an additional postdoc, Dr. Satoe Swai-Toyota, has been hired to replace Dr. Choi. An undergraduate, Amanda Miller, is also working on this project for independent study credits.		
COI Name (Institution):	Masson, Patrick (University of Wisconsin-Madison) Spalding, Edgar (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX09AK80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research seeks to define the intra-/inter-cellular mechanisms behind the generation and transmission of the gravity signal. It is widely accepted that sedimentation of starch filled amyloplasts in the columella of the root cap and endodermis of the hypocotyl act to initiate the gravitropic signal. However, although pH and Ca2+-based signals have been proposed as key elements in this plant response, the precise mechanisms translating physical force to biochemically transmissible signal is still poorly defined. Needed is a high resolution spatiotemporal map of changes in ion activities induced by the gravity signal. Such mapping will be achieved by confocal imaging of plants expressing a range of GFP-based probes targeted to different subcellular locales within the Arabidopsis seedling. In parallel, we will use machine-vision technologies to quantify the distribution of curvature along the root and hypocotyl axes during gravitropism. Curvature and ion change distribution maps will be measured in wild type and mutants known to be defective in the pathways that lead to gravity signal transduction. By mapping the detailed spatial and temporal dynamics of ion signaling to these morphometric analyses, we will be able to pinpoint where and when each component is altering tropic response. Potential for translation to future spaceflight opportunities: These studies will develop a morphometric technology highly applicable to spaceflight analysis where existing hardware can be used to collect the time-lapse imaging required for morphometric analysis. Such analysis will develop a quantitative description of the effects of the flight environment on plant development at levels of architecture ranging from branching patterns to the kinematics of growth. This will allow us to ask questions only addressable by the unique environment of Space; for example, defining the effects of mutations on gravitational response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has capitalized on the development of automated image analyses algorithms that allow for the automated extraction of quantitative parameters of growth such as growth rate, organ angle, rate of curvature, and high resolution kinematic analyses of growth rates along the surface of the responding roots. Although these are being applied to root gravitropism in Arabidopsis, they hold the potential to be applicable to a much wider set of growth responses, allowing for the rapid screening of plants for quantitative differences in their growth patterns. The work is also developing the use of calcium-sensitive green fluorescent protein-based probes that have very high affinity for calcium. These are allowing for the visualization of calcium signals that were previously undetectable and has applications to a wide range of stress signaling pathways in plants.
Task Progress:	This grant covers research to map the dynamics of ionic signaling associated with gravity sensing and response onto the kinetics of gravitropic growth. We are using this analysis to assess how well such signaling events can be linked to the control of plant development. Over the last year we have been able to: (1) image Ca2+ signaling events in wild type plants and in a range of mutant backgrounds, (2) apply high-resolution kinematic analyses of cell-level growth in these plants, and (3) characterize the potential for cross-talk between anoxic and gravitropic signaling and response. (1) Ca2+ signaling events in the gravitropic response. We have previously used plants transformed with the GFP-based Ca2+ sensitive protein cameleon YC3.6 to image the Ca2+ changes occurring upon gravistimulation (applied by rotation of the plant through 90 degrees) using confocal imaging. This sensor allowed us to resolve a major Ca2+ change occurring 5-10 minutes after gravistimulation. This change is asymmetrical, being localized to the lower face of the root cap and oscillates with a 5 minute periodicity. The timing of these changes suggest that they play a role in assymetrical auxin transport in the root tip. However, 'YC-nanos', Ca2+ reporters with much higher sensitivities (i.e. able to resolve much lower amplitude Ca2+ increases), are now available and we have generated plants expressing a range of these sensors, including YC-nano140 and YC-nano65. Using these probes, we have been able to monitor an array of Ca2+ changes that were previously undetected. These Ca2+ changes occur in the root cap columella (site of gravity perception) and in the lateral root cap cells within 2 minutes of gravistimulation. This time-frame is within the presentation time for Arabidopsis, i.e. it is within the timing predicted for the generation of gravity signal transduction in the root cap. Prominent Ca2+ increases also occur on the lower side of the root cap in lateral root cap cells. After 10 minutes, the changes seen with YC3.6 were also observed with these new sensors, suggesting that the reason we see many more Ca2+ changes with the YC-nanos is their superior sensitivity. We have therefore transformed these new Ca2+ sensors into plants where the gravitropic signaling is predicted to be disrupted. Our initial analyses indicate such Ca2+ signaling is altered in the pgm1-1 and adg1-1 mutants. These mutants are defective in starch synthesis in the root cap. This work is beginning to suggest a relationship between Ca2+ signaling in the root cap and the starch statolith mechanism of sedimentable starch-filled amyloplasts that is predicted to underlie initial gravisensing events in plants. We have also been collaborating with the lab of Janet Braam (Rice University) on imaging Ca2+ signaling in calmodulin-like protein mutants that likely form part of the response network for Ca2+ signaling changes. We have determined that one such mutant (cml24) has effects on root growth, leading to root skewing, cell file rotation and altered microtubule orientation, suggesting a model whereby Ca2+ and CML24 act to control microtubule function and so contribute to the control of root directional growth. Although CML24 mutants are affected in root growth and root touch responses, they show no lesion in root gravitropism and no clear alteration in Ca2+ signals as monitored by YC3.6. (2) Kinematic analyses of gravitropic responses. In collaboration with Dr. Nathan Miller in the Spalding lab, we have been applying software that allows the automated analysis of regional growth rates over the surface of the root. The analysis program is helping define when and where on the surface of the root gravitropic growth responses are detectable. We are using this mapping to drive where we are concentrating on analyzing the dynamics of Ca2+ responses after the initial gravisignaling events (that occur over the first 5 minutes) have been generated and the cell elongation phase of gravity response is initiated. In wild-type plants we have monitored an extended Ca2+ increase on the lower side of the root at this time period coinciding with the initiation of asymmetrical growth. Using this approach, we have also mapped responses in the pgm1-1 and adg1-1 mutants. Although the pgm1-1 and adg1-1 mutants are knockouts for these two starch biosynthetic genes, their agravitropic phenotypes are actually conditional, being rescued by supplementation of sugars to the medium. The sugars likely bypass these mutations and facilitate starch deposition in the root cap. Our kinematic analyses of these mutants under conditions where they show reduced gravitropism has pinpointed where the local reductions in regional growth that underlie the slower bending response of the roots occur.

	We are currently using mutant lines expressing YC-nano140 and YC-nano65 to ask how the Ca2+ signaling in these roots is altered and whether this correlates with these sites of reduced tropic response. (3) Role of anoxia on root gravitropic growth responses. During our analysis of gravitropic growth, we have become increasingly aware that root zone hypoxia/anoxia alters the kinetics of the response. In addition, we have also characterized, using our YC-nano-expressing plants, that anoxia due to poor oxygenation of growth media leads to a complex Ca2+ signaling response in the root tip, especially in the columella cells, the cells responsible for gravity sensing. Our mining of the publicly available transcriptional databases suggests that Ca2+ response proteins such as the calcium-dependent protein kinases, calmodulin-like proteins and Ca2+ transporters of the ACA and CAX family are highly over-represented in datasets from anoxia-treated plants. At a practical level, we now know that anoxic response can trigger Ca2+ changes in the root that may confound analysis of gravitropic signaling and so we take measures to ensure normoxic conditions for the root zone in our experiments. However, we also hypothesize that anoxic Ca2+ signaling may be involved in rewiring the Ca2+ response machinery to gravitropic Ca2+ signals. Unexpectedly, we have found that mutants in ACA1 and CAX2 are both resistant to anoxia and show reduced gravitropic response suggesting they may play roles in the Ca2+ signaling seen in both processes. Kinematic analyses of the regional growth rates of these mutants is being performed to map the cellular basis for their reduced bending response. We are currently generating aca1 and cax2 lines expressing the YC-nano sensors to ask how these affect gravitropism-related Ca2+ signaling. Root zone hypoxia is predicted to be an important issue when growing plants under microgravity as buoyancy-driven convective gas mixing is lost in these conditions and so oxygen supply to the root becomes limiting. Thus, these interactions between anoxia and tropic signaling may have important implications for optimally growing plants in the spaceflight environment.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Wang Y, Wang B, Gilroy S, Chehab EW, Braam J. "CML24 is involved in root mechanoresponses and cortical microtubule orientation in Arabidopsis." Journal of Plant Growth Regulation 2011 Dec;30(4):467-79. http://dx.doi.org/10.1007/s00344-011-9209-9 , Dec-2011
Articles in Peer-reviewed Journals	Choi W-G, Swanson SJ, Gilroy S. "High-resolution imaging of Ca2+ , redox status, ROS and pH using GFP biosensors." The Plant Journal. 2012 Apr;70(1):118-28. http://dx.doi.org/10.1111/j.1365-313X.2012.04917.x ; PubMed PMID: 22449047 , Apr-2012
Books/Book Chapters	Swanson SJ, Gilroy S. "Calcium: from root macronutrient to mechanical signal." in "Plant Roots: The Hidden Half. 4th edition. " Ed. A. Eshel, T. Beekman. Danvers, Mass. : CRC Press, in press. March 2012., Mar-2012
Books/Book Chapters	Choi W-G, Swanson SJ, Gilroy S. "Calcium, mechanical signaling, and tip growth." in "Coding and Decoding of Calcium Signals in Plants." Ed. S. Luan. New York, NY : Springer, 2011. p. 41-61. Signaling and communication in plants, vol. 10. http://dx.doi.org/10.1007/978-3-642-20829-4 , Aug-2011
Books/Book Chapters	Swanson SJ, Gilroy S. "Imaging changes in cytoplasmic calcium using the Yellow Cameleon 3.6 biosensor and confocal microscopy." in "Plant Lipid Signaling Protocols." Ed. T. Munnik, I Heilmann. Humana Press, in press. As of March 2012; expected publication June 2012., Mar-2012